

Could the weak force pick life's hand in an icy-moon ocean?

DRAFT — not submitted; for student + mentor review; every number traces to README.md or EXPERIMENT.md

Abstract

Life uses left-handed amino acids and right-handed sugars, and no one knows why. One old idea (Kondepudi & Nelson 1985) is that the weak nuclear force makes one hand very slightly lower in energy (the parity-violating energy difference, PVED), and that a large, slow, self-amplifying chemical system could turn that tiny bias into a one-handed outcome. We ask when the bias beats random chance, using stochastic simulations checked against a formula, and apply the answer to the subsurface oceans of Enceladus and Europa and to early Earth's ocean.

A rule of thumb comes out (§3.0): about 10^{33} – 10^{34} molecules, a few billion moles, must take part in the decision *together*. That means kilometre-scale volumes of well-mixed water at micromolar to millimolar concentrations. Ponds and single rock pores are many orders of magnitude too small. Icy-moon oceans are big enough, so the answer turns on how well they mix. Their salinity sets how strongly they are stratified, and a stratified ocean mixes slowly up and down. The patchwork of hands then freezes into stacked layers that merge only as fast as vertical mixing allows (§3.3; pictures in Fig. 2). Varying every uncertain input at once, physics wins in 22% (Enceladus), 29% (Europa) and 75% (early Earth) of plausible cases (§3.9).

The racemic amino acids in samples returned from asteroid Bennu, plausibly a fragment of a wet parent body, are a strong test (§3.10). Every amplifier in our assumed rate range would also have run on Bennu's parent body. If an amplified excess lasted forever, Bennu would rule out nearly all of them and cut physics' odds to 2–15%. But amplification is a non-equilibrium process. In a closed rock whose heat runs out, the same reaction network (run with every step reversible) rises to a large excess and then falls back to 50/50. In an ocean that vents keep driving, the excess holds. With that rule, a racemic Bennu fits the model, and the odds stay at 22% / 29% / 75% (12–25% / 17–31% / 42–75% across the cases tested). The idea makes a testable prediction: amino acids from a driven ocean, such as Enceladus' plume, should carry a lasting excess, while amplified excesses of molecules that can racemize should fade in extinct closed parent bodies. Meteorites fit this: their surviving excesses are in amino acids that cannot racemize or that locked into crystals (§3.10). Bennu's racemic isovaline, which cannot racemize, is not explained by the rule and remains open. Connected compartments, such as the pores of a vent mound, help only in proportion to the volume they connect within the decision time (§3.11). Long runs show no late drift or flipping at realistic molecule numbers (§3.12). No measured prebiotic reaction yet amplifies handedness this way, though a recently proposed network built from two measured reactions could (Higgs & Blackmond 2025), so we pre-register a wet-lab search sized for meteoritic concentrations (≤ 1 mM) and published measurement precision (§5). All code is public and self-checking (§6).

1 Introduction

Almost all life uses left-handed (“L”) amino acids in proteins and right-handed (“D”) sugars in DNA/RNA, even though both mirror-image forms are chemically identical alone. One idea, from Kondepudi & Nelson (1985), is that the weak force makes left- and right-handed molecules very slightly different in energy (PVED). That bias is astronomically small, but a large enough, slow enough, self-amplifying chemical system (each hand helps make more of itself while suppressing the other) could in principle let it beat chance.

No one had applied this size-and-time test to icy-moon oceans before this project; a scan of all 435 papers citing Kondepudi & Nelson 1985 (the paper that proposed the idea) or Brandenburg & Multamäki 2004 (closest prior spatial work, no weak force) found none that do (README, Novelty check). The question: is an icy-moon or early-Earth ocean large and mixed enough for the weak-force bias to reliably beat chance, or is the outcome decided by luck? This is a student computational project. Every number below is either produced and self-checked by a script in the project folder, or a cited literature number (flagged where used), with the source README “Part” noted in brackets.

2 Methods

2.1 The core model (`sim.py`)

Let $\alpha = (L - D)/(L + D)$ be the excess of one hand. Near the point where a self-amplifying reaction “switches on”, any such reaction reduces to the same simple equation (the normal form; Fig. 1):

$$d\alpha = (\lambda(t)\alpha - \alpha^3 + g)dt + \sqrt{\varepsilon} dW, \quad \lambda = \gamma t.$$

- λ is how far past the switch-on point the system is. Before it ($\lambda < 0$) the only stable state is 50/50; after it ($\lambda > 0$) there are two stable states, all-L and all-D.
- $g = \Delta E_{\text{pv}} / kT$ is the weak-force bias (ΔE_{pv} is the PVED per molecule; its sign for real amino acids is unsettled).
- $\varepsilon \approx 1/N$ is the noise from counting individual molecules, N being the number of molecules that react together.
- $\gamma = 1/(k\tau)$ is how quickly conditions sweep through the switch-on point, where $k = k_2c$ is the self-copying rate (rate constant k_2 times concentration c) and τ is the time the sweep takes. Time is measured in units of $1/k$.

Linearising around $\lambda = 0$ gives the chance that the weak-force-favoured hand wins:

$$P = \Phi(\Delta), \quad \Delta = \sqrt{2} \pi^{1/4} g \gamma^{-1/4} \varepsilon^{-1/2} = \sqrt{2} \pi^{1/4} g (k\tau)^{1/4} \sqrt{N},$$

where Φ is the normal cumulative distribution. $\Delta = 2$ means a 97.7% win. The formula is checked against 4,000 brute-force runs per point (agreement ± 0.005 , Part 1).

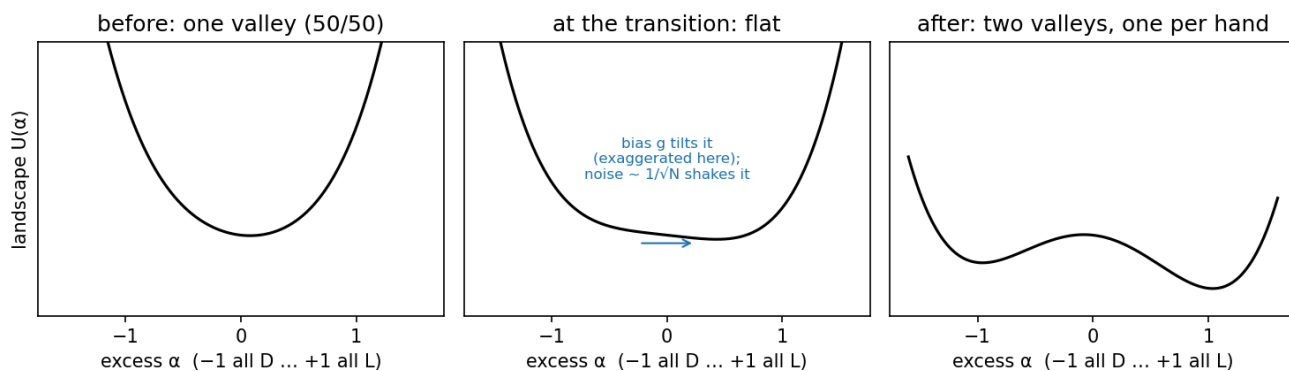


Fig. 1. The landscape the excess α moves in. The weak force tilts it very slightly; molecular noise shakes it. Which valley α falls into is decided in a short window around the switch-on point.

2.2 Oceans (`ocean.py`, `scenarios.py`, `uncertainty.py`)

For a body of water, $N = c \cdot V \cdot f \cdot 1000 \cdot N_A$, where V is the volume (m^3), f the fraction that is well mixed on the decision time, and N_A Avogadro's number. Physics “wins” only if two conditions both hold: the size condition $\Delta \geq 2$, and the time condition $k_2\tau \geq 10$ (the reaction actually finishes). `uncertainty.py` draws 200,000 input sets per world, log-uniform over each plausible range: bias g , rate k_2 , concentration c , horizontal and vertical mixing, time, volume, and the real-chemistry factor F (§2.4).

2.3 Space and mixing (`domains.py`, `domains3d.py`, `coarsen3d.py`, `k3.py`, `shell3d.py`, `aniso3d.py`, `sphere_layers.py`)

With mixing, α becomes a field $\alpha(x, t)$:

$$\partial_t \alpha = \lambda \alpha - \alpha^3 + g + D_h \nabla_h^2 \alpha + D_z \partial_z^2 \alpha + \text{noise}.$$

D_h and D_z are the horizontal and vertical eddy diffusivities. Rescaling depth by $\sqrt{(D_h/D_z)}$ turns this exactly into the equal-mixing equation in a box that is taller by that factor. So weak vertical mixing acts like a deep, narrow ocean. After the decision, walls between regions of opposite hand move at $D \times \text{curvature}$ (Allen & Cahn 1979). Closed patches shrink, so patch size grows as $l = A\sqrt{(Dt)}$, and the majority hand takes over (Fig. 2). Flat walls between horizontal layers do not move.

Numerical scheme. All spatial runs use explicit Euler (Euler–Maruyama when noise is included), which is first order in time, together with second-order centred differences (the 7-point Laplacian) in space. Walls are periodic sideways; top and bottom are closed (no flux). Units are $1/k$ for time and $\sqrt{(D/k)}$ for length, with grid spacing 1 and time step 0.05. That satisfies the 3D stability limit $D \cdot dt/dx^2 \leq 1/6$. The resolution and time-step checks are in §3.12.

2.4 Real chemistry (`frank.py`, `frank2.py`)

The normal form is checked against a full Frank-type network in a flow reactor. Here A is the achiral feedstock (for example an amino-acid precursor), and L and D are the two hands of the product:

reaction	rate	role
feed $\rightarrow$ A	$f(a_0 - a)$, a_0 ramped slowly upward	supply; the ramp sweeps the system through switch-on
$A \rightarrow L$, $A \rightarrow D$	k_{oa} each	slow uncatalysed production (unbiased)
$A + L \rightarrow 2L$	$k(1 + g)aL$	self-copying; the weak force enters as $\pm g$
$A + D \rightarrow 2D$	$k(1 - g)aD$	
$L + D \rightarrow$ waste	$k_i LD$	mutual destruction (“cross-inhibition”)
$L, D \rightarrow$ out	$k_d L, k_d D$	outflow

The two ingredients that make it amplify are self-copying and mutual destruction. Every reaction is simulated with its own counting noise (the chemical Langevin equation). At the switch-on point the network maps onto the normal form: $g_{\text{eff}} = 2gxka$, $\varepsilon = (2k_{oa} + 2x(ka + k_d))/\Omega$, and $\gamma = k \cdot da/dt$, where x is the concentration of each hand and Ω the reactor size. The extra noise from non-copying reactions lowers Δ by a factor $F \approx 0.7$ compared with §2.2. `frank2.py` adds reverse reactions as a second scheme.

2.5 Other scripts

- `inherit.py`: a pool that inherits a small excess (Part 4b).
- `dual.py`: coupled amino-acid and sugar systems (Part 5).
- `scorecard.py`: real reactions scored against seven requirements (Part 6).
- `design.py`: experiment sizing (§5).
- `review.py`: the checks added after expert feedback (§3.0, §3.10–3.12, §5).
- `coarsen_fig.py`, `sketch.py`: the figures.
- `shell_res.py`: the grid-resolution test.

Every script asserts its own checks.

3 Results

3.0 A common-sense rule: how many molecules must decide together? (`review.py` R1)

Setting $\Delta = 2$ in the formula gives the number of molecules that must react as one well-mixed pool:

$N \geq \left(2/(\sqrt{2} \pi^{1/4} g (k\tau)^{1/4})\right)^2$. For $g = 10^{-17}$ (the middle of the modern PVED range) and a reaction that only just finishes ($k\tau = 10$), $N \geq 3.6 \times 10^{33}$ molecules, or 6×10^9 mol. At 1 μM that is a well-mixed cube of water 18 km on a side; at 1 mM, 1.8 km; at 0.1 M, 0.4 km. A 1 m³ pond at 1 M holds only 6×10^{26} molecules, about 7 orders of magnitude too few.

The PVED is a fixed energy, so the bias $g = \Delta E/kT$ is larger in the cold. The needed N scales as T^2 : 3.6×10^{33} at 0 °C, 4.3×10^{33} at 27 °C, 6.7×10^{33} at 100 °C. Cold helps only by a factor of about 2. Time helps only weakly, as $(k\tau)^{1/4}$: a sweep 10,000× slower lowers N by 100×. The practical message is that concentration alone cannot rescue a small setting. Only ocean-scale water that is mixed within the decision time can pool enough molecules.

3.1 Is an icy-moon ocean big enough? (Part 1)

At a representative bias $g = 1e-17$ and rate $k_2 = 1e-3$ /M/s, the minimum reactant concentration needed for physics to win is ~ 2 mM for a small lake, but only ~ 50 pM for Enceladus, ~ 4 pM for Earth, and ~ 1 pM for Europa. Size is not the limit for icy-moon oceans across the modern PVED range. Time becomes the limit when chemistry is slow: at $k_2 = 1e-6$ /M/s the time condition sets the answer for every ocean (~ 3 nM for Enceladus); temperature jitter barely matters, shifting the win probability by < 0.01 . If only a fraction f of the ocean is well mixed, the effective bias scales as $g \cdot \sqrt{f}$ — 1% mixing on Enceladus behaves like a bias ten times smaller.

3.2 One ocean, or a patchwork? (Part 2)

Adding 1D spatial mixing, three laws were derived and checked to a few percent. A single ocean-wide outcome needs $D \geq 3 \times 10^{-6}$ m²/s (Enceladus) or 1×10^{-5} m²/s (Europa). Molecular diffusion ($\sim 10^{-9}$ m²/s) is far too slow for either, but the mixing rate modeled by Zeng & Jansen (2021), 5×10^{-5} m²/s, clears both. With only diffusion, oceans fragment into km-scale patches and the favoured hand gets just 50–58% of Enceladus (up to 79% of Europa at 1 μ M) — a mostly chance-decided, mixed-hand ocean; with modelled mixing, physics wins ocean-wide down to ~ 1 nM (Enceladus) or ~ 1 pM (Europa).

3.3 The 3D correction — patches heal themselves (Part 3)

In full 3D, the 1D patch-size law fails (31% spread). Curved 3D patch walls straighten themselves out, so small patches shrink even with no bias: size grows as $l = A \cdot \sqrt{(D \cdot t)}$, exponents 0.45/0.49 (theory 0.5), $A = 5.48/5.60$ at two mixing strengths (a 2026-09-23 correction fixed a rounded “5.5 at both”; conclusion unchanged). Starting from a 55% majority, it grows to 66% ($t=50$), 81% ($t=200$), 93% ($t=400$) — standard phase-ordering physics. Healing within the moon’s age needs $D \geq 6 \times 10^{-6}$ m²/s (Enceladus) or 2×10^{-5} m²/s (Europa); diffusion alone would take 6×10^{11} – 2×10^{13} years, modelled mixing heals in $\sim 1 \times 10^7$ (Enceladus) or $\sim 4 \times 10^8$ years (Europa). *Correction (2026-09-25)*: the thresholds above apply one mixing value over the ocean’s width, and the 10^{-10} – 10^{-3} m²/s range (and the 5×10^{-5} “modelled” value) are Zeng & Jansen’s **vertical** diffusivity (their Sec. II.2). Treating the directions separately: sideways healing needs only $\geq \sim 6 \times 10^{-4}$ (Enceladus) or $\sim 2 \times 10^{-4}$ m²/s (Europa), about 180–450 $\times$ below the ~ 0.1 m²/s sideways eddy mixing estimated by Zhang, Kang & Marshall (2024) (50–140 $\times$ below 0.03, the low end of their runs), so it never limits. Healing over the depth (≈ 40 km Enceladus, ≈ 120 km Europa) needs vertical mixing $\geq \sim 2 \times 10^{-9}$ – 2×10^{-6} m²/s (Enceladus) or $\sim 4 \times 10^{-9}$ – 2×10^{-7} m²/s (Europa), which still sits inside the vertical range: *Update (same day, aniso3d.py)*: that top-to-bottom rule is the wrong picture for most of the range. Weak vertical mixing is exactly equivalent to equal mixing in a taller ocean (stretch depth by $\sqrt{(D_h/D_z)}$); the stretched ocean is deeper than wide for 92–96% of the plausible range. In such tall boxes the patchwork froze into 3–5 stacked layers in 7/8 runs (unchanged from $t = 500$ to 1000), while a cube healed to one hand in 4/4 runs from a 55% start. On a round moon, layer boundaries are spheres: a wavy boundary was measured to flatten at $D_h \cdot k^2$ regardless of D_z (0.00967 vs 0.00964 and 0.00239 vs 0.00241), which holds in a flat box. *Correction (2026-09-26)*: on a sphere “down” follows the radius, so a constant-depth boundary feels only vertical mixing and sinks at $2 \cdot D_z/R$, not $2D_h/R$ as first written (which gave takeover within $\leq 1.5 \times 10^4$ yr Enceladus, $\leq 2.8 \times 10^5$ yr Europa). `sphere_layers.py` measured sink speeds of $0.991/0.991/0.984 \times 2 \cdot D_z/r$ with 0.0% change when $D_h \times 4$; takeover now takes up to 1.5×10^{12} yr (Enceladus) or 2.8×10^{13} yr (Europa) at $D_z = 10^{-10}$ m²/s, and the merge-in-time condition holds in 33%/34% of draws (MEASURED). The hemisphere split is still unstable (rate $1.017 \times$ predicted; an off-equator band vanished), because of a shift off the great circle (a tilt is just a rotation, so cannot matter). The top layer decides with its own molecules (a share $L/H' = 0.25$ of the ocean,

ASSUMED; a column-like sphere run measured 0.52 and 0.07 in 2 runs, neither supporting nor ruling it out); the favoured hand ended on top in 3/4 runs from 55% vs 2/4 from 50% (MEASURED, small sample). A direct per-patch check (k3.py) found an effective decision volume of 60.2 cells, consistent to 1% across three bias strengths (a correction fixed a rounded “≈57 cells”). Three more seeds gave $K_3 = 3.698, 3.481, 3.652$ (original 3.642), a 2.3% spread across all four (MEASURED). Fig. 2 shows both behaviours: a cube healing to the majority, and a tall (weakly mixed) box freezing into layers that do not change from $t = 400$ to $t = 5000$.

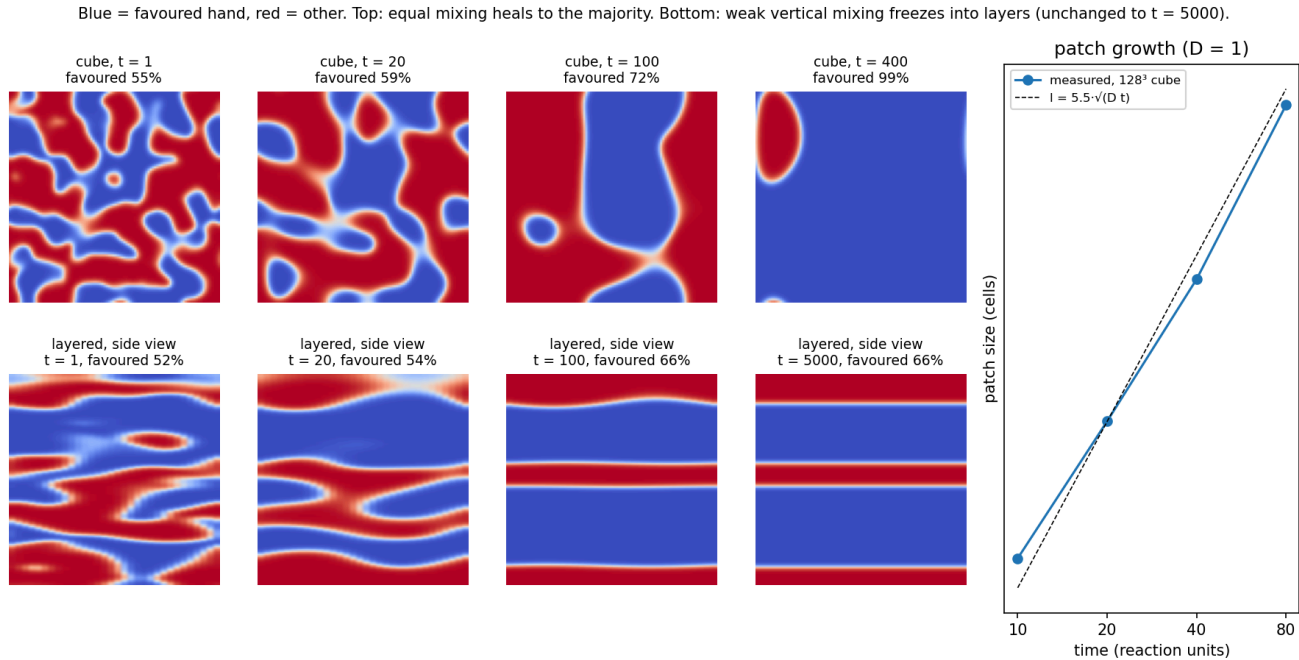


Fig. 2. Coarsening. Top: slices of a 96^3 cube starting from a 55% majority; patches round off and shrink, and the majority reaches 99% by $t = 400$. Bottom: side view of a tall box (equivalent to weak vertical mixing); the patchwork freezes into flat layers of opposite hands, unchanged to $t = 5000$. Right: patch size grows as $l = 5.5\sqrt{(Dt)}$ (exponent 0.45). `coarsen_fig.py`.

For early Earth ($1.33 \times 10^{18} \text{ m}^3$, mean depth 3.7 km), a one-ocean outcome needs $D \geq 5 \times 10^{-4} - 2 \times 10^{-2} \text{ m}^2/\text{s}$ — cleared by horizontal eddies ($\sim 10^3 \text{ m}^2/\text{s}$, Abernathy & Marshall 2013) and even by vertical mixing alone ($1.1 \times 10^{-5} \text{ m}^2/\text{s}$, Ledwell et al. 1993: 94% win at 1 nM, 100% at 1 μM). Earth’s own left-handedness can’t confirm any of this, though — all life shares one ancestor, so any mechanism (chance included) ends with one hand; a second, independent ocean (Enceladus, Europa) is the real test.

3.4 Where was the hand decided? (Part 4)

Open ocean gives physics ~100% odds; lake/lagoon 51–100%; warm little pond 50–52% (coin flip); vent pore 50% (coin flip). The tension: settings chemists favour for concentrating molecules (ponds, vent pores) are exactly where physics loses, since they hold too few molecules total. The concentration inputs are the weakest part of this: Stribling & Miller (1987)’s $\sim 3 \times 10^{-4} \text{ M}$ early-ocean amino-acid estimate is the only sourced number, with no lower estimate found in the literature — treat it as contested, not consensus.

3.5 Can small pools inherit a hand? (Part 4b)

A pool that fills with water already carrying a small excess can reliably follow it above a threshold, checked against 4,000 runs/case across 5 cases. Required inherited excess: 3×10^{-14} (lake/lagoon), 6×10^{-12} (pond), 1×10^{-6} (vent

pore). The weak force alone supplies only $\sim 5 \times 10^{-18}$ at equilibrium — far too small — but an ocean that already amplified the bias can supply up to ~ 1 , and meteorites (Murchison L-isovaline, Glavin & Dworkin 2009, 18.5% excess) supply enough by 5–13 orders of magnitude. This made meteorite seeding look like the simplest explanation, though isovaline is not itself a protein amino acid and its excess would need catalytic pass-through (shown for isovaline $\rightarrow$ D-sugar, Pizzarello & Weber 2004). *Update*: samples returned from Bennu contain chiral non-protein amino acids that are racemic or nearly so (Glavin et al. 2025), and Ryugu’s are also racemic (Furusho et al. 2024). The Bennu authors conclude that delivered molecules may not explain life’s handedness. So meteoritic excesses are not universal; they vary with parent-body history (Elsila et al. 2016; Glavin et al. 2020). One caveat is that the excess a pond needs (6×10^{-12}) is some 10 orders of magnitude below what any lab can measure (about $\pm 1\%$). “Racemic within error” therefore neither supplies nor rules out a pond-sized seed.

3.6 Why L-amino acids pair with D-sugars (Part 5)

Two coupled chiral systems, each helping the partner’s hand, were checked against 4,000 runs/case. Real chemistry links them both ways: L-amino acids catalyse D-sugar excess (Pizzarello & Weber 2004; Breslow & Cheng 2010); D-RNA prefers L-amino acids $\sim 4\times$ (Tamura & Schimmel 2004). Without coupling the hands match only 50% of the time; with $\kappa \approx \sqrt{\gamma}$, 99.9%. The winning pair depends on the *sum* of the two biases — predicted/simulated probabilities of L-amino-acid + D-sugar ranged from 0.500 (biases cancel) to 0.800 (biases add) across four tested combinations. Coupling means physics only makes one combined decision — but the sugar PVED sign is unsettled, and if it actually favours L-sugars (opposite natural D), the biases would partly cancel, pushing back toward a coin flip. A 2005 whole-DNA-helix calculation (Faglioni et al.) concluded the weak force does *not* favour nature’s helices, per its abstract. “Not the natural helix” is weaker than a strong push toward the mirror helix; the full paper was inaccessible (paywalled), so this is checked against the abstract only.

3.7 The missing reaction (Part 6)

Seven requirements for a self-amplifying prebiotic chiral reaction were derived (self-copying; mutual suppression; driven away from 50/50; works in water from simple molecules; fast enough; works from a tiny excess; biologically relevant product). Scoring seven real candidates, the best (Noorduyn 2008 attrition deracemization) scores 6.0/7, but **no known system meets self-copying + suppression + prebiotic plausibility + relevant product together**. Speed is never the limiting factor — natural chemistry has 10^5 – 10^8 years, so even extremely slow reactions qualify. This gap motivates the wet-lab experiment in Section 5.

3.8 Does real chemistry follow the formula? (Part 7)

The network (§2.4, table) is the classic Frank (1953) scheme: an achiral feedstock A turns into L or D, each hand copies itself from A, and the two hands destroy each other on contact. Self-copying makes whichever hand leads grow faster, and mutual destruction removes the minority. Together they turn a small lead into a one-handed outcome. The network, simulated with molecular counting noise from every reaction step, follows the simplified formula within ~ 1 percentage point in the slow (natural) limit (2.5 points near saturation); a fast sweep selects *more* strongly than predicted, so the formula is conservative. The real network’s selection strength is lower by a conversion factor $F \approx 0.7$, because every reaction adding/removing a chiral molecule adds noise, not just self-copying ones — so every concentration in Parts 1–4 should rise by about $1.6\times$. No conclusion changes, since margins there were $10\times$ – $10^6\times$, except the Europa mixing margin (not concentration-based).

3.9 Robustness checks (Part 8)

Varying every uncertain input at once (200,000 draws/world), physics wins in **75%** of plausible cases for early Earth's open ocean, **22%** for Enceladus and **29%** for Europa on a round moon (3% and 7% if layers never merged); the merge-in-time condition holds in 33%/34% of draws. Vertical mixing is the most decisive input (0%/0%/52% win rate in the low half of its range vs. 43%/57%/97% in the high half, Enceladus/Europa/Earth); concentration (+21/+10) and time (+15/+11; +32 for Earth) come next, then bias g (+8/+4) and rate k_2 (+5); faster sideways mixing lowers the rate slightly (-2 to -9); F and volume ≤ 2 for the moons. (*Corrections 2026-09-25: first reported as 18% for both moons, from a run that used the vertical-mixing range for sideways healing; then 43%/58% before layering was modelled. Correction 2026-09-26: then 100%/55%/78% with concentration deciding (21%/57% vs. 89%/99%), bias g +18 to +23, k_2 +14, vertical mixing +10 to +14, from a run that let layer boundaries sink at $2 \cdot D_h/R$ instead of $2 \cdot D_z/R$.)* A second reaction scheme with back-reactions matches the formula within 1.7 points and gives $F = 0.71$ (vs. 0.70). A thin, wide "shell" shape closer to a real ocean's proportions still shows the majority taking over, just more slowly (growth exponent 0.41 vs. 0.45–0.49 in a cube).

3.10 Reconciling the model with racemic Bennu and Ryugu (review.py R2)

Bennu is plausibly a fragment of a wet parent body. Its chiral amino acids, including isovaline, which cannot racemize after it forms, are racemic or nearly so (Glavin et al. 2025). This is a direct test of the model, because the model assumes that some amplifying chemistry exists. The inputs below are ASSUMED ranges: amino acids at 10–330 nmol/g of rock (330 nmol/g is the highest value for LAP 02342, from J. Dworkin, pers. comm.), a water-to-rock mass ratio of 0.1–1 (giving 10^{-5} to 3×10^{-3} M in the fluid), and aqueous alteration lasting 1–10 Myr. Then $c \cdot \tau$ on Bennu's parent body was 3×10^8 to 9×10^{11} M·s. Any amplifier with $k_2 \geq 10^{-11}$ to 3×10^{-8} /M/s would have finished there. With only pore-water diffusion, the finished patches would be 1–3 km across, so each gram-sized sample would sit inside one patch and show a large excess of one hand or the other. The racemic samples say no such amplifier finished there.

Suppose the same chemistry, with the same rate constant, operated on Bennu's parent body and in the icy-moon oceans:

- **Rates in the original range (10^{-6} to 1 /M/s):** every draw would have finished on Bennu. Taken at face value, Bennu rules out the whole range.
- **Rates extended down to 10^{-12} /M/s:** 23% of draws keep Bennu racemic. Among those, physics wins in only **2% (Enceladus), 4% (Europa) and 15% (early Earth)** of cases. An amplifier slow enough to stay idle on Bennu is usually too slow to finish in an icy moon as well. Only 1–5% of icy-moon draws have a larger $c \cdot \tau$ than Bennu's maximum.

Taken at face value this is the strongest constraint in the paper. But it assumes that an amplified excess, once made, lasts. That is only true while something keeps driving the system.

Closed rock vs open ocean (review.py R6). Amplification is a non-equilibrium process. At equilibrium, both hands are equal (apart from the 10^{-17} PVED), so a closed system must end up racemic. To check this, we took the frank2.py network, made every step reversible with rates that obey detailed balance, and ran it closed (no feed, no outflow). Starting from a 0.1% seed:

- with reverse steps about 1/2000 the speed of forward ones, the excess peaks at 70% and is back to 50/50 after about 7×10^3 reaction times ($1/k_2c$);
- with reverse steps 100× slower, it reaches 100% and fades after about 6×10^7 reaction times;
- the same network kept open (fed and flushed) holds 99.8% for as long as it is driven.

Bennu’s parent body was closed, and its liquid water lasted only while short-lived radioactive heat did. Its excess therefore stays at zero if the reaction never finished ($k_2c\tau < 10$) or if it finished and then faded ($k_2c\tau$ above the fade time). The second branch removes most of the conflict:

k₂ range	fade time (reaction times)	P(win given a racemic Bennu): Enceladus / Europa / Earth
$10^{-6} - 1$	7×10^3	22% / 29% / 75% (unchanged)
$10^{-6} - 1$	6×10^7	25% / 31% / 75%
$10^{-12} - 1$	7×10^3	15% / 21% / 57%
$10^{-12} - 1$	6×10^7	12% / 17% / 42%

This resolution has two assumptions. First, an icy-moon ocean must be continuously driven; Enceladus’ measured plume H₂ points to ongoing water–rock reactions (Waite et al. 2017). H₂ is a reactive fuel, so seeing it today means it is still being made; whether that supply has been steady over long times is not known (Glein & Zolotov 2020). The evidence leans toward a long-lived drive. Tidal heating of a porous core can keep hydrothermal activity going for tens of Myr to Gyr (Choblet et al. 2017). Serpentinization alone can supply the observed H₂ for at most about 500 Myr (Daval et al. 2022), but radiolysis can run for Gyr, and tidal grinding of rock could add bursts lasting up to Myr (Öberg, Magnabosco & Tosca 2026, preprint). Even 500 Myr is far longer than the decision times in this model. Second, the chemical drive on Bennu’s parent body must have stopped while liquid water remained. There is reasonable evidence that Bennu is the remnant of an ocean world (J. Dworkin, pers. comm.), so “closed” here cannot mean “had no ocean”. It means the ocean stopped being fed. The distinction that matters is how long the drive lasts compared with the water. On the parent body, both the heat (from short-lived ²⁶Al) and the fresh rock for water–rock reactions ran out within about 10 Myr, and Bennu’s material is almost completely altered, which suggests the rock reactions ran to completion before the water was gone. Enceladus is still tidally heated and still producing H₂ today. If instead the drive and the water ended together, any excess would have frozen in, and for those cases the conflict returns. Bennu’s veins and evaporite minerals show that its fluid moved and dried out, so the timing is open and is the main question this section leaves. The resolution also makes a prediction that can be checked: samples from a driven ocean, such as Enceladus’ plume, should carry a lasting excess, while amplified excesses of molecules that can racemize should fade in extinct, closed parent bodies. A racemic Enceladus would count against the idea. Capillary-electrophoresis instruments being built for ocean worlds can separate enantiomers in salty samples (Mora et al. 2022), so this is a practical measurement.

What the meteorites say about this rule. Thermodynamics already requires it: a lasting excess needs a constant supply of free energy, and only open systems can keep one (Blackmond 2004; Plasson et al. 2007). The meteorites fit if the rule is read as applying to excesses that stay dissolved and can racemize. In Murchison and Murray, the α-H amino acids, which racemize by losing their α-hydrogen, show no excess, and the excesses survive only in α-methyl amino acids such as isovaline, which cannot racemize (Pizzarello & Cronin 2000). The large L-aspartic and L-glutamic excesses in Tagish Lake (43–59%) are in amino acids that crystallize as separate L and D crystals, while alanine, which cannot, is racemic. Glavin et al. (2012) explain them as amplification during

alteration with the excess locked into solids, which takes it out of the reversible solution chemistry the rule describes. Two points remain open. First, isovaline cannot racemize, so the fade rule does not explain Bennu’s racemic isovaline. It says instead that Bennu’s parent body either received no starting bias in isovaline or never amplified one, which is a statement about its history and not about this model. Second, the Murchison isovaline excess grows with alteration (Glavin & Dworkin 2009), so some amplification did happen in a closed body. The rule only says such an excess cannot last in solution, and solid or non-racemizable products escape that. If the rule is still set aside, the Bennu conflict returns and physics’ odds fall back to 2%/4%/15%. Both results are reported.

If the closed-system argument fails, the other escape routes all say that the chemistry differed:

- the amplifier needed something Bennu’s fluid lacked (longer peptides, mineral surfaces, warmer or longer-lived water);
- Bennu’s water was mostly pore water in mud, not a free ocean;
- the amino-acid concentration in Bennu’s fluid is not the concentration of the amplifier’s reactants.

Each escape route is testable, but none is shown here. Temperature differences between the two settings are not modelled.

3.11 Compartments: can a network of connected pools beat one pond? (review.py R3)

Several authors suggest that life began in connected compartments, such as the pores of a hydrothermal mound (Milner-White & Russell 2005; Russell, *Scientific American*), acting together as one “reactor”. We simulated 16 compartments, each with 1/16 of the molecules, exchanging material at rate q (normal form, $g = 10^{-3}$, $\gamma = 0.01$):

exchange rate q × decision time	P(majority favoured)	runs ending all one hand
0 (isolated)	0.675	0%
0.1	0.686	0%
1	0.724	53%
10	0.714	100%

A single pool holding all the molecules gives $P = 0.724$, and one isolated compartment gives 0.559. So compartments that exchange faster than the decision time ($\sqrt{(\tau/k)}$) act as one pool of their combined volume, and afterwards they heal to one hand. Isolated compartments vote by majority but stay a mix of hands forever.

In a real vent mound ($k_2 = 10^{-3}$ /M/s, $c = 1$ mM, 10^5 yr), the decision takes about 56 years. Pores 1 mm apart exchange by diffusion 2×10^6 times faster than that, so they are fully connected. Diffusion reaches only about 1.3 m in 56 years, however, so a diffusion-connected mound behaves like a ~ 2 m³ pond: $P = 0.500$, a coin flip. If fluid flow connects a 100 m mound (10^6 m³), $P = 0.73$; a flow-connected cubic kilometre gives $P = 1.000$. Compartments do beat ponds, but only by the volume that fluid flow links together within the decision time. That makes a flowing vent system a serious candidate, consistent with §3.0.

3.12 Long runs and numerical resolution (review.py R4, coarsen_fig.py, shell_res.py)

Does the decided hand drift later? We continued the sweep with the reaction held at saturation for $100\times$ longer than the decision itself. The share of runs on the favoured hand was 0.730 at the end of the sweep, 0.730 after 10^3 more time units, and 0.730 after 10^4 . Once decided, a pool flips only by a rare noise-driven jump. We measured

the flip rate at high noise and found it matches Kramers' rate $(\sqrt{2/2\pi}) \cdot e^{(-1/2\varepsilon)}$ within 5%: 1.8×10^{-2} , 7.7×10^{-3} , 3.4×10^{-3} and 1.4×10^{-3} per unit time at $\varepsilon = 0.2, 0.15, 0.12$ and 0.1 . With $\varepsilon = 1/N$ the rate falls as $e^{(-N/2)}$. A $1 \mu\text{m}^3$ droplet at $1 \mu\text{M}$ (about 600 molecules) can flip; a pond or ocean never does. The frozen layers in Fig. 2 show no change between $t = 400$ and $t = 5000$ (a $12\times$ longer run).

Grid resolution (suggested by A. Brandenburg). The thin-shell runs (§3.9) use 8 grid points in depth, and a wall is only about 1.4 cells wide. We reran the same physical box ($256 \times 256 \times 8$ units) with grid spacing 1 ($256 \times 256 \times 8$ points) and $1/2$ ($512 \times 512 \times 16$ points), with the time step scaled by dx^2 . A run with grid spacing 1 and a $4\times$ smaller time step gave identical patch sizes (30, 37, 47, 68, 91 at $t = 25-400$), so the time step is converged. On the fine grid, patch sizes were 31, 39, 48, 64 and 93, within 5% of the coarse grid at every time, and the growth exponent was 0.39 against 0.41. The coarse grid therefore resolves the coarsening, and the 8-point depth does not change the results; this is expected because the scheme is second order in space. We also ran a smaller box ($128 \times 128 \times 8$ units) with 2 seeds per case to check majority takeover. The coarse and fine grids agree in direction (a 55% majority grows on both), but runs from exact 50/50 vary too much between seeds to say more (`shell_res.py`).

4 Limitations

- All results are order-of-magnitude estimates; the $F \approx 0.7$ real-chemistry conversion (Part 7) is checked for only two reaction schemes; other schemes could give a smaller F .
- Part 2's spatial laws are 1D; Part 3's 3D correction uses finite 128^3 boxes, and the 55%→93% late-time takeover is shown only qualitatively (patches reach the box edge).
- Real ocean turbulence isn't simple diffusion; icy-moon vertical mixing (Zeng & Jansen 2021) is a model input, uncertain across 10^{-10} – $10^{-3} \text{ m}^2/\text{s}$ — the single biggest reason the icy-moon answer isn't settled (Parts 2–3). Sideways mixing ($\sim 0.1 \text{ m}^2/\text{s}$, Zhang et al. 2024) never limits healing. On a round moon layers merge only as fast as vertical mixing allows (checked on a spherical shell, `sphere_layers.py`), so vertical mixing is again the key unknown; the deciding top layer's thickness is an estimate.
- τ (how slowly conditions change) is set to the ocean's age, the most favourable case possible.
- No measured prebiotic reaction performs the needed self-amplifying chirality; the Soai reaction does it in the lab but isn't prebiotic. The closest prebiotic candidates are peptide-ligation reactions that amplify an excess (Deng, Yu & Blackmond 2024) and a model network combining them with amino acid synthesis that breaks symmetry on its own (Higgs & Blackmond 2025). That model also finds the excess lasts only in an open, fed system and returns to racemic in a closed one, the same result as `review.py` R6. Its rates are given as ratios, so k_2 is still a swept assumption, not measured (Part 6).
- PVED has never been measured directly; its sign for amino acids in water is unsettled (the “conformation problem”, MacDermott et al. 2009a; that work also finds the L-zwitterion of alanine PVED-stabilized in its lowest-energy conformation), though the L-forms of the Murchison α -methyl amino acids are calculated to be PVED-stabilized (MacDermott et al. 2009b), and the sugar PVED sign is also unsettled, with some evidence pointing the “wrong” way, which would partly cancel the amino-acid bias (Part 5).
- **Molecules flipping hand.** A chiral molecule can in principle tunnel to its mirror form. For amino acids the barrier makes this far slower than the age of the universe, and the PVED is larger than the tunnelling splitting, so each molecule stays in one hand (Quack 2002). The flipping that does happen is chemical: racemization by loss and return of the α -hydrogen. It is random and goes both ways at the same rate apart from the PVED tilt, so it adds noise rather than a bias, and it is the reverse-reaction route behind the closed-system fade (§3.10). A simulation that adds random flipping at rate r to the model (`review.py` R7) confirms

this: the chance the weak force wins only drifts toward 50% (72% → 67% as r goes from 0 to 0.45 of the amplifier rate), and if flipping outpaces amplification no hand is chosen at all.

- **A possible second weak-force bias (beta decay).** Beta decay is a weak-force process that emits electrons spinning mostly one way. Vester, Ulbricht & Krauch (1959) proposed that these destroy one hand faster than the other. In the lab, low-energy spin-polarized electrons break up one enantiomer of bromocamphor more often than the other, by a few parts in 10^4 (Dreiling & Gay 2014). An icy moon's rocky core contains radioactive ^{40}K , U and Th, so its water receives such electrons. The model does not include this. It would add to g , with the same sign in every ocean, but its sign for amino acids and its size in an ocean (dose, shielding by water, fraction of molecules hit) are unknown, so it is listed as a possible extra bias and not used. A simulation (review.py R8) shows what it would take: with an asymmetry of 3×10^{-4} , the beta effect matches the PVED once about one molecule in 10^{13} is destroyed by polarized electrons per reaction time. For an ocean at the edge (PVED alone gives 84%), that raises the odds to 98% if beta pushes the same way, or cuts them to 50% if it pushes the other way.
- The Enceladus ocean volume ($2.7 \times 10^{16} \text{ m}^3$) is derived indirectly from ice-shell/core geometry (Čadek et al. 2016), self-checked by ocean.py against that paper's implied range ($2.45\text{--}2.93 \times 10^{16} \text{ m}^3$); the ocean's age is separately debated (1 Myr–1 Gyr).
- Physics wins for Enceladus/Europa in 22%/29% of the plausible input range, early Earth 75% (Part 8) — not settled, and it falls to 3%/7% if layers never merge; vertical mixing is the key unknown. *History: first reported as 18%, then 43%/58% (corrected 2026-09-25), then 55%/78% (and 100% for Earth) with concentration as the key unknown; corrected 2026-09-26 because layer boundaries on a sphere sink at $2 \cdot D_z/R$, not $2 \cdot D_h/R$ (sphere_layers.py).*
- **Bennu (§3.10).** If an amplified excess lasted forever, Bennu's racemic samples would cut physics' win rate to 2%/4%/15%. The closed-rock result removes most of that conflict, but it depends on Bennu's parent body behaving as a closed system and on its reverse reaction rates, which are unmeasured. Meteorite excesses survive only in non-racemizable or crystallized amino acids, which fits the rule, but it does not cover Bennu's isovaline, which cannot racemize (§3.10). Both the 22%/29%/75% and the 2%/4%/15% results should be treated as possible. All Bennu inputs (concentration, duration, same k_2 in both settings) are assumed.
- Concentrations: meteoritic amino acids reach at most a few hundred nmol/g (J. Dworkin, pers. comm.), or $\leq \sim 1 \text{ mM}$ in parent-body fluid. The pond (10^{-2} M) and vent-pore (up to 1 M) values in Part 4 are upper bounds. Lowering them only strengthens the conclusion that small settings are coin flips.
- The mixing model uses simple eddy diffusion. Salinity-driven stratification enters only through the assumed vertical-diffusivity range (10^{-10} to $10^{-3} \text{ m}^2/\text{s}$); no ocean circulation is simulated.
- Meteorite seeding (Part 4b) predicts the same left-handed outcome throughout the Solar System as the weak-force hypothesis — finding left-handed life on an icy moon can't by itself distinguish the two; only life from another star system could.

5 Planned experiment

Because no known prebiotic reaction satisfies the Part 6 requirements, EXPERIMENT.md pre-registers a wet-lab search for one; the pre-registration was written before any data were collected. Hypothesis: at least one prebiotic, water-based peptide-forming system will show seed-following growth of enantiomeric excess (ee).

Revised after expert feedback (J. Dworkin). The first design used 0.1 M and assumed 0.2% ee precision. Both were unrealistic:

- Meteoritic amino acids reach at most a few hundred nmol/g, or $\leq \sim 1$ mM in parent-body fluid.
- Published ee uncertainties are ± 0.01 –1.5% by GC-MS and 1.2–7.2% by LC-MS. Glavin & Dworkin (2009) reported 2.6% from repeat measurements.

The smallest detectable rate after one year (review, py R5) is $k_{2,\min} = \ln(1 + 3\sqrt{2} \sigma/e_0)/(cT)$, where σ is the ee noise per sample, e_0 the seed excess, c the concentration and T the run length:

concentration	seed	$\sigma = 0.2\%$	$\sigma = 1\%$	$\sigma = 2.6\%$	2.6%, 9 replicates
1 mM	5%	5×10^{-6}	2×10^{-5}	4×10^{-5}	2×10^{-5}
1 mM	20%	1×10^{-6}	6×10^{-6}	1×10^{-5}	5×10^{-6}
0.1 M	5%	5×10^{-8}	2×10^{-7}	4×10^{-7}	2×10^{-7}

At realistic concentrations the pond floor is 3×10^{-6} /M/s (1 mM, 100 yr), and the ocean floor is 3×10^{-9} (1 μ M, 10^8 yr). No lab run can reach the ocean floor at natural concentration. The revised design therefore has two tiers:

1. **Realistic arm:** 1 mM, a 20% seed (similar to Murchison isovaline), 3 years, and 9 replicate measurements at 2.6%. This reaches 1.8×10^{-6} /M/s, below the pond floor, and asks directly whether amplification happens at meteoritic concentrations.
2. **Accelerated arm:** 0.01 M and 0.1 M, stated plainly as *not* natural conditions. It exists to find any amplifier at all and to measure how the ee growth rate scales with concentration. That scaling (the rate law) is what the model needs in order to extrapolate to 1 μ M.

Four candidate chemistries are tested, chosen from the Part 6 scorecard:

- A. amino acids + carbonyl sulfide (Leman, Orgel & Ghadiri 2004);
- B. amino acids + hydroxy acids under wet-dry cycling (Forsythe et al. 2015);
- C. cysteine peptides catalysing their own joining (Foden et al. 2020). This is the priority arm.
- D. an RNA precursor on magnetite (Ozturk et al. 2023).

A Viedma (2005) grinding positive control checks that the pipeline can detect real amplification.

Each arm gets +L, +D (mirror control) and racemic seedings in triplicate, plus these negative controls:

- no activator;
- sterile-filtered;
- seed alone, to measure racemization.

Analysts are blinded. An arm counts as an amplifier only if:

- both seeded sets rise by more than the detection threshold;
- the L- and D-seeded sets change by equal and opposite amounts;
- the racemic vials stay at zero;
- the controls stay flat.

The D-seed mirror arm is the main safeguard against contamination by biological (L) amino acids. Following Dworkin's advice, the realistic arm would be proposed to local groups with chiral GC-MS (NASA Ames: G. Cooper, G. Chaban; San José State: A. Rios) rather than as funded outside work.

As of this draft, this experiment has not been run.

6 Reproducibility

All computational results were produced by the Python scripts in the project folder, each printing and asserting its own self-checks (numpy, scipy and matplotlib are required). The code is public at

<https://github.com/helloamonkey742-code/chirality-project> (MIT licence) so that every number can be reproduced:

```
./run_all.sh          # fast subset, ~3 minutes
./run_all.sh --full   # adds the slow 3D and real-chemistry runs
```

Individual scripts (sim.py, ocean.py, domains.py, patches.py, domains3d.py, coarsen3d.py, scenarios.py, dual.py, inherit.py, scorecard.py, design.py, k3.py, frank.py, frank2.py, shell3d.py, aniso3d.py, sphere_layers.py, uncertainty.py, review.py, coarsen_fig.py, shell_res.py) can also be run one at a time; approximate runtimes are listed in README.md's "Run" section.

References

- Kondepudi & Nelson 1985, *Nature* 314:438, doi:10.1038/314438a0
- Quack 2002, "How important is parity violation for molecular and biomolecular chirality?", *Angew. Chem. Int. Ed.* 41:4618, doi:10.1002/anie.200290005
- MacDermott, Fu, Hyde, Nakatsuka et al. 2009, "Electroweak parity-violating energy shifts of amino acids: the conformation problem", *OLEB* 39:407, doi:10.1007/s11084-009-9161-x (2009a)
- MacDermott et al. 2009, "Parity-violating energy shifts of Murchison L-amino acids are consistent with an electroweak origin of meteorite L-enantiomeric excesses", *OLEB* 39:459, doi:10.1007/s11084-009-9162-9 (2009b)
- Chandrasekhar 2008, "Molecular homochirality and the parity-violating energy difference. A critique with new proposals", *Chirality* 20:84, doi:10.1002/chir.20502
- Brandenburg 2019, "The limited roles of autocatalysis and enantiomeric cross-inhibition in achieving homochirality in dilute systems", *OLEB* 49:49, doi:10.1007/s11084-019-09579-4
- Hochberg, Buhse, Micheau & Ribó 2023, "Chiral selectivity vs. noise in spontaneous mirror symmetry breaking", *PCCP* 25:31583, doi:10.1039/d3cp03311b
- Čadek et al. 2016, *Geophysical Research Letters* 43, doi:10.1002/2016GL068634
- Postberg et al. 2018, *Nature*, doi:10.1038/s41586-018-0246-4
- Anderson et al. 1998, *Science* 281:2019
- Brandenburg & Multamäki 2004, *Int. J. Astrobiology*, doi:10.1017/s1473550404001983
- Sandars 2005, *Int. J. Astrobiol.*, doi:10.1017/s1473550405002338
- Gleiser & Walker 2008, "Punctuated chirality", *OLEB*, doi:10.1007/s11084-008-9147-0
- Jafarpour, Biancalani & Goldenfeld 2017, *PRE* 95:032407, doi:10.1103/PhysRevE.95.032407
- Zeng & Jansen 2021, *PSJ* 2:151, doi:10.3847/PSJ/ac1114
- Zhang, Kang & Marshall 2024, *Science Advances* 10, doi:10.1126/sciadv.adn6857
- Allen & Cahn 1979, *Acta Metallurgica* 27:1085, doi:10.1016/0001-6160(79)90196-2 (boundaries move at $D \times \text{curvature}$)

- Charette & Smith 2010, *Oceanography* 23(2), doi:10.5670/oceanog.2010.09
- Ledwell, Watson & Law 1993, *Nature* 364:701, doi:10.1038/364701a0
- Abernathey & Marshall 2013, *JGR Oceans* 118:901
- Bray 1994, “Theory of phase-ordering kinetics”, *Adv. Phys.* 43:357
- Stribling & Miller 1987, *OLEB* 17:261, doi:10.1007/BF02386466
- Pearce et al. 2017, *PNAS* 114:11327, doi:10.1073/pnas.1710339114
- Baaske et al. 2007, *PNAS* 104:9346, doi:10.1073/pnas.0609592104
- Toner & Catling 2020, *PNAS* 117:883, doi:10.1073/pnas.1916109117
- Pizzarello & Weber 2004, *Science* 303:1151, doi:10.1126/science.1093057
- Breslow & Cheng 2010, *PNAS* 107:5723, doi:10.1073/pnas.1001639107
- Hein, Tse & Blackmond 2011, *Nat. Chem.* 3:704, doi:10.1038/nchem.1108
- Tamura & Schimmel 2004, *Science* 305:1253, doi:10.1126/science.1099141
- Ozturk et al. 2023, *Sci. Adv.* 9:eadg8274, doi:10.1126/sciadv.adg8274
- Faglioni, D’Agostino, Cadioli & Lazzeretti 2005, “Parity violation energy of biomolecules II: DNA”, *Chem. Phys. Lett.* 407:522, doi:10.1016/j.cplett.2005.04.009
- Glavin & Dworkin 2009, *PNAS* 106:5487, doi:10.1073/pnas.0811618106
- Frank 1953, *Biochim. Biophys. Acta* 11:459
- Soai et al. 1995, *Nature* 378:767, doi:10.1038/378767a0; Sato et al. 2003, *Angew. Chem.* 42:315, doi:10.1002/anie.200390105
- Viedma 2005, *PRL* 94:065504, doi:10.1103/PhysRevLett.94.065504
- Noorduyn et al. 2008, *Angew. Chem.* 47:6445, doi:10.1002/anie.200801846
- Saghatelian et al. 2001, *Nature* 409:797, doi:10.1038/35057238
- Joyce et al. 1984, *Nature* 310:602, doi:10.1038/310602a0
- Klussmann et al. 2006, *Nature* 441:621, doi:10.1038/nature04780
- Leman, Orgel & Ghadiri 2004, *Science*, doi:10.1126/science.1102722
- Forsythe et al. 2015, *Angew. Chem.*, doi:10.1002/anie.201503792
- Foden et al. 2020, *Science*, doi:10.1126/science.abd5680
- Glavin et al. 2025, “Abundant ammonia and nitrogen-rich soluble organic matter in samples from asteroid (101955) Bennu”, *Nature Astronomy*, doi:10.1038/s41550-024-02472-9
- Furusho et al. 2024, “Enantioselective three-dimensional HPLC determination of amino acids in the Hayabusa2 returned samples from the asteroid Ryugu”, *J. Chromatogr. Open*, doi:10.1016/j.jcoa.2024.100134
- Glavin et al. 2020, “The search for chiral asymmetry as a potential biosignature in our Solar System”, *Chem. Rev.* 120:4660, doi:10.1021/acs.chemrev.9b00474
- Elsilá et al. 2016, “Meteoritic amino acids: diversity in compositions reflects parent body histories”, *ACS Cent. Sci.* 2:370, doi:10.1021/acscentsci.6b00074
- Glavin et al. 2012, “Unusual nonterrestrial L-proteinogenic amino acid excesses in the Tagish Lake meteorite”, *Meteorit. Planet. Sci.* 47:1347, doi:10.1111/j.1945-5100.2012.01400.x
- Milner-White & Russell 2005, *Orig. Life Evol. Biosph.* (compartments; suggested by A. Brandenburg; not yet read in full)
- Kramers 1940, *Physica* 7:284 (escape rate over a barrier)
- Waite et al. 2017, “Cassini finds molecular hydrogen in the Enceladus plume: evidence for hydrothermal processes”, *Science* 356:155, doi:10.1126/science.aai8703

- Glein & Zolotov 2020, “Hydrogen, hydrocarbons, and habitability across the Solar System”, *Elements* 16:47–52, doi:10.2138/gselements.16.1.47
- Mora, Kok, Noell & Willis 2022, “Detection of biosignatures by capillary electrophoresis mass spectrometry in the presence of salts relevant to ocean worlds missions”, *Astrobiology* 22:914, doi:10.1089/ast.2021.0091
- Blackmond 2004, “Asymmetric autocatalysis and its implications for the origin of homochirality”, *PNAS* 101:5732, doi:10.1073/pnas.0308363101 (closed systems and equilibrium)
- Vester, Ulbricht & Krauch 1959, “Optische Aktivität und die Paritätsverletzung im β -Zerfall”, *Naturwissenschaften* 46:68, doi:10.1007/BF00599091
- Dreiling & Gay 2014, “Chirally sensitive electron-induced molecular breakup and the Vester-Ulbricht hypothesis”, *Phys. Rev. Lett.* 113:118103, doi:10.1103/PhysRevLett.113.118103
- Higgs & Blackmond 2025, “Autocatalytic symmetry breaking and chiral amplification in a feedback network combining amino acid synthesis and ligation”, *PNAS* 122:e2423683122, doi:10.1073/pnas.2423683122
- Deng, Yu & Blackmond 2024, “Symmetry breaking and chiral amplification in prebiotic ligation reactions”, *Nature* 626:1019, doi:10.1038/s41586-024-07059-y
- Choblet et al. 2017, “Powering prolonged hydrothermal activity inside Enceladus”, *Nature Astronomy* 1:841, doi:10.1038/s41550-017-0289-8
- Daval, Choblet, Sotin & Guyot 2022, “Theoretical considerations on the characteristic timescales of hydrogen generation by serpentinization reactions on Enceladus”, *JGR Planets* 127, doi:10.1029/2021JE006995
- Öberg, Magnabosco & Tosca 2026, “Tidal rock grinding as a source of H₂ on Enceladus”, arXiv:2606.16860 (preprint)
- Plasson, Kondepudi, Bersini, Commeyras & Asakura 2007, “Emergence of homochirality in far-from-equilibrium systems: mechanisms and role in prebiotic chemistry”, *Chirality* 19:589, doi:10.1002/chir.20440
- Pizzarello & Cronin 2000, “Non-racemic amino acids in the Murray and Murchison meteorites”, *Geochim. Cosmochim. Acta* 64:329, doi:10.1016/S0016-7037(99)00280-X